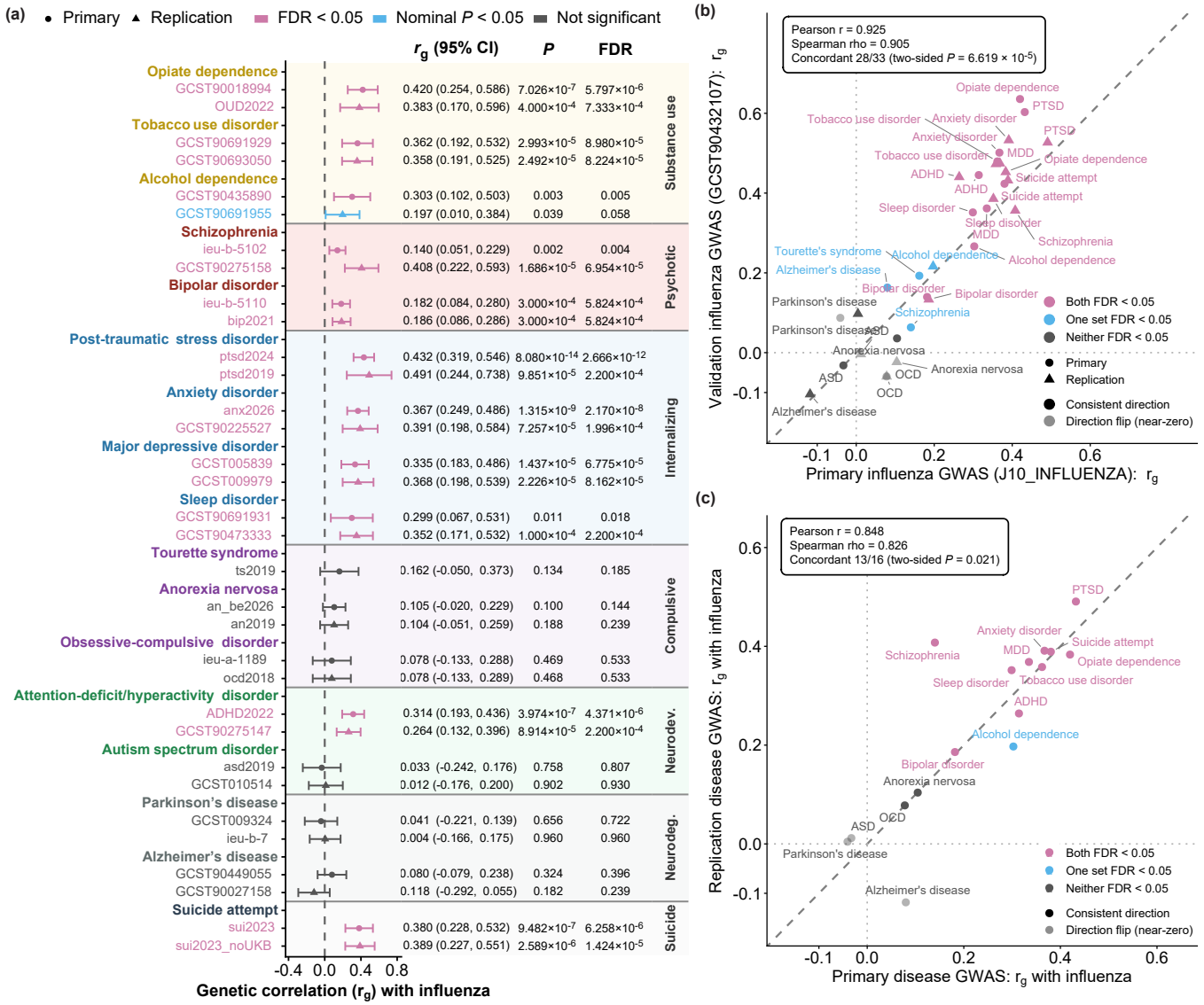

(a)
Primary
Replication
FDR < 0.05
Nominal P < 0.05
Not significant
rg (95% CI)
P
FDR
Opiate dependence
GCST90018994 0.420 (0.254, 0.586) 7.026×10-7 5.797×10-6
OUD2022 0.383 (0.170, 0.596) 4.000×10-4 7.333×10-4
Tobacco use disorder
GCST90691929 0.362 (0.192, 0.532) 2.993×10-5 8.980×10-5
GCST90693050 0.358 (0.191, 0.525) 2.492×10-5 8.224×10-5
Alcohol dependence
GCST90435890 0.303 (0.102, 0.503) 0.003 0.005
GCST90691955 0.197 (0.010, 0.384) 0.039 0.058
Substance use
Schizophrenia
ieu-b-5102 0.140 (0.051, 0.229) 0.002 0.004
GCST90275158 0.408 (0.222, 0.593) 1.686×10-5 6.954×10-5
Bipolar disorder
ieu-b-5110 0.182 (0.084, 0.280) 3.000×10-4 5.824×10-4
bip2021 0.186 (0.086, 0.286) 3.000×10-4 5.824×10-4
Psychotic
Post-traumatic stress disorder
ptsd2024 0.432 (0.319, 0.546) 8.080×10-14 2.666×10-12
ptsd2019 0.491 (0.244, 0.738) 9.851×10-5 2.200×10-4
Anxiety disorder
anx2026 0.367 (0.249, 0.486) 1.315×10-9 2.170×10-8
GCST90225527 0.391 (0.198, 0.584) 7.257×10-5 1.996×10-4
Major depressive disorder
GCST005839 0.335 (0.183, 0.486) 1.437×10-5 6.775×10-5
GCST009979 0.368 (0.198, 0.539) 2.226×10-5 8.162×10-5
Sleep disorder
GCST90691931 0.299 (0.067, 0.531) 0.011 0.018
GCST90473333 0.352 (0.171, 0.532) 1.000×10-4 2.200×10-4
Internalizing
Tourette syndrome
ts2019 0.162 (-0.050, 0.373) 0.134 0.185
Anorexia nervosa
an_be2026 0.105 (-0.020, 0.229) 0.100 0.144
an2019 0.104 (-0.051, 0.259) 0.188 0.239
Obsessive-compulsive disorder
ieu-a-1189 0.078 (-0.133, 0.288) 0.469 0.533
ocd2018 0.078 (-0.133, 0.289) 0.468 0.533
Compulsive
Attention-deficit/hyperactivity disorder
ADHD2022 0.314 (0.193, 0.436) 3.974×10-7 4.371×10-6
GCST90275147 0.264 (0.132, 0.396) 8.914×10-5 2.200×10-4
Autism spectrum disorder
asd2019 0.033 (-0.242, 0.176) 0.758 0.807
GCST010514 0.012 (-0.176, 0.200) 0.902 0.930
Neurodev.
Parkinson's disease
GCST009324 0.041 (-0.221, 0.139) 0.656 0.722
ieu-b-7 0.004 (-0.166, 0.175) 0.960 0.960
Alzheimer's disease
GCST90449055 0.080 (-0.079, 0.238) 0.324 0.396
GCST90027158 0.118 (-0.292, 0.055) 0.182 0.239
Neurodeg.
Suicide attempt
sui2023 0.380 (0.228, 0.532) 9.482×10-7 6.258×10-6
sui2023_noUKB 0.389 (0.227, 0.551) 2.589×10-6 1.424×10-5
Suicide
-0.4 0.0 0.4 0.8
Genetic correlation (rg) with influenza
(b)
Pearson r = 0.925
Spearman rho = 0.905
Concordant 28/33 (two-sided P = 6.619 × 10-5)
Validation influenza GWAS (GCST90432107): rg
Primary influenza GWAS (J10_INFLUENZA): rg
Both FDR < 0.05
One set FDR < 0.05
Neither FDR < 0.05
Primary
Replication
Consistent direction
Direction flip (near-zero)
(c)
Pearson r = 0.848
Spearman rho = 0.826
Concordant 13/16 (two-sided P = 0.021)
Replication disease GWAS: rg with influenza
Primary disease GWAS: rg with influenza
Both FDR < 0.05
One set FDR < 0.05
Neither FDR < 0.05
Consistent direction
Direction flip (near-zero)